

Discuss one thing you notice that is different between the two emails that might relate to the identification of spam.

The ham email looks very legit, while the spam email is parsed in HTML. The links are explicit in ham, but in the spam, the link is hidden/hyperlinked behind a href. Spam might be more related to faulty looking links or hidden links from this first impression of spam/ham.

0.0.1 Question 3

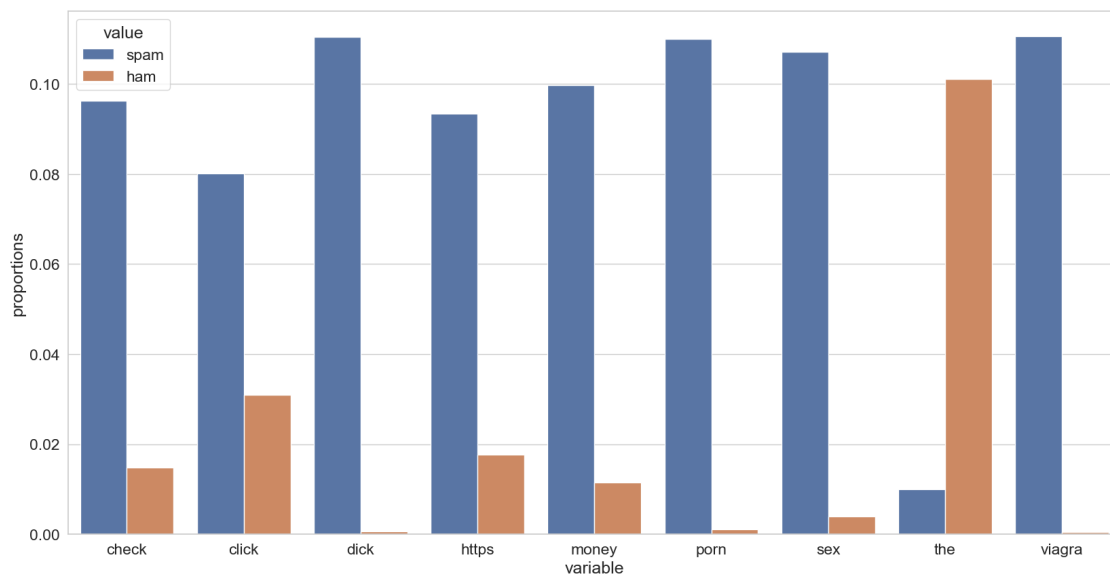
Create a bar chart like the one above comparing the proportion of spam and ham emails containing certain words. Choose a set of words that are different from the ones above, but also have different proportions for the two classes. Make sure to only consider emails from `train`.

```
In [11]: train=train.reset_index(drop=True) # We must do this in order to preserve the ordering of email
```

```
words = ['porn', 'sex', 'dick', 'viagra', 'money', 'click', 'check', 'the', 'https']

data = words_in_texts(words, train['email'])
df = pd.DataFrame(data=data, columns=words)
df = pd.concat([train['spam'], df], axis=1)
df = df.melt('spam')
df = df['value'].groupby(df['variable']).value_counts() / len(df['value'])
df = df.to_frame()
df.columns = ["proportions"]
df = df.reset_index()
df['value'] = df['value'].replace([0, 1], ["spam", "ham"])
plt.figure(figsize=(20, 10))
sns.barplot(x="variable", y="proportions", data=df, hue="value")
```

```
Out[11]: <AxesSubplot:xlabel='variable', ylabel='proportions'>
```



0.0.2 Question 6c

Provide brief explanations of the results from 6a and 6b. Why do we observe each of these values (FP, FN, accuracy, recall)?

There are no false positives, since we label everything as ham. There are however false negatives since we labeled everything as spam (1), so we are bound to have some false negatives. Accuracy is relatively high, mostly because we have an imbalance of data labels. Recall is also defined as 0, since true positives aren't available as we never classify spam (1).

0.0.3 Question 6e

Are there more false positives or false negatives when using the logistic regression classifier from Question 5?

There are more false negatives than false positives. There are 122 data points labeled as false positives and 1699 data points labeled as false negatives.

0.0.4 Question 6f

1. Our logistic regression classifier got 75.76% prediction accuracy (number of correct predictions / total). How does this compare with predicting 0 for every email?
 2. Given the word features we gave you above, name one reason this classifier is performing poorly. Hint: Think about how prevalent these words are in the email set.
 3. Which of these two classifiers would you prefer for a spam filter and why? Describe your reasoning and relate it to at least one of the evaluation metrics you have computed so far.
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1. Our zero classifier has 74.47% accuracy. Our logistic regression model barely makes any improvement.
 2. One reason why classifier is performing poorly is that we have uneven distribution of labels. There are far more labels associated with ham rather than spam. So maybe to improve our model, we need to get more spam data points, or choose features that have better distinction between ham and spam.
 3. I would take constant model. It's only slightly worse than the logistic regression model. There are no false positives, so I reduce the risk of having an important ham email missing my inbox because it gets misclassified as spam.

